1. Hypothesis testing is a statistical method which entails formulating a specific hypothesis about the population and then using sample evidence to decide whether to accept or reject the hypothesis. We take an example of hypothesis tests concerning the population mean (Does μ, the population mean of Y, equal a specific value, μ_0). The first step is to specify the hypothesis to be tested, called the null hypothesis:

$$H_0 : \mu = \mu_0. \tag{1}$$

The alternative hypothesis specifies what would happen if the null hypothesis is not true. The (two-sided) alternative hypothesis is

$$H_1 : \mu \neq \mu_0. \tag{2}$$

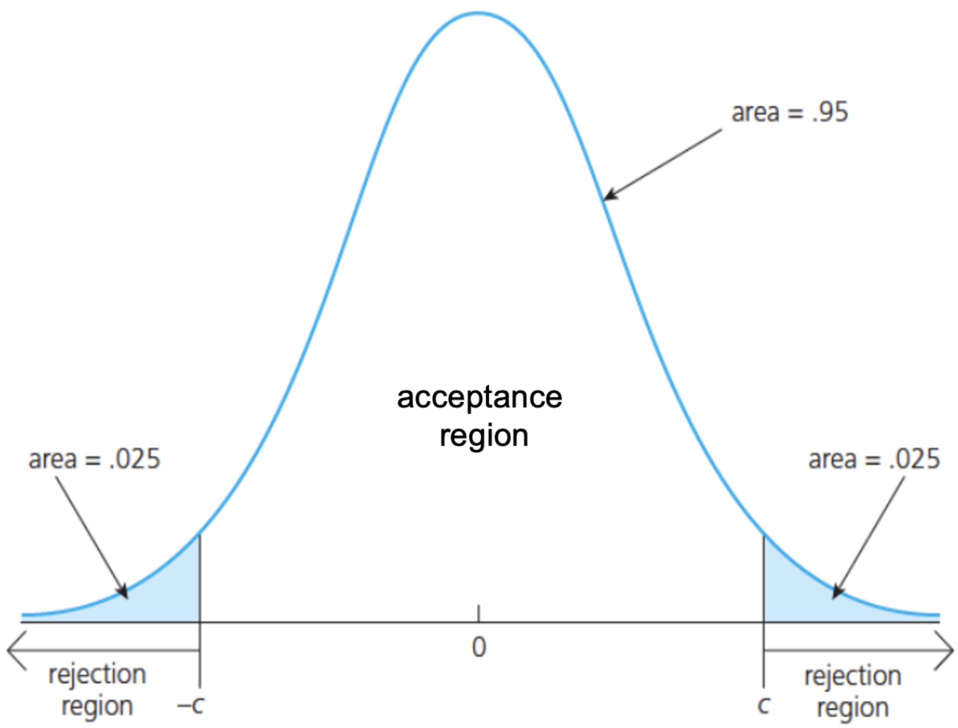
To perform the hypothesis testing, a test statistics is required. In our example, we use the t-statistic:

$$t = \frac{\bar{Y} - \mu_0}{SE(\bar{Y})}, \tag{3}$$

where $\bar{Y}$ is the sample average and $SE(\bar{Y})$ is the standard error of $\bar{Y}$.

The null distribution represents the distribution of the test statistic if the null hypothesis is true, while the alternative distribution reflects the distribution of the test statistic when the alternative hypothesis is true. In the current example, if the null hypothesis is true, the t-statistic follows $N(0, 1)$ in large samples (we focus on the large-sample distribution).

The critical value of the test statistic is the threshold at which the null hypothesis is just rejected at a specified significance level, such as $\alpha = 0.05$. The rejection region is the range of values for the test statistic that lead to rejecting the null hypothesis, while the acceptance region includes the range of values for which the null hypothesis is not rejected. The following diagram shows the distribution of the test statistic under null hypothesis, along with the acceptance and rejection regions for a two-sided test at $\alpha = 0.05$, and c is the critical value.



2. (a) In large samples (when the OLS assumptions are satisfied), $\hat{\beta}_1$ follows the normal distribution centered at the true value β_1 (since $\hat{\beta}_1$ is an unbiased estimator of β_1):

$$\hat{\beta}_1 \sim N\left(\beta_1, \sigma^2_{\hat{\beta}_1}\right), \tag{4}$$

where $\sigma^2_{\hat{\beta}_1}$ denotes the variance of $\hat{\beta}_1$ under large samples.
(In this course, we always focus on the large-sample distributions (e.g., the normal distribution) instead of small-sample distributions.)

(b) If we want to test $H_0 : \beta_1 = 1$, the corresponding t-test statistic is:

$$t = \frac{\hat{\beta}_1 - 1}{SE(\hat{\beta}_1)}. \tag{5}$$

Therefore, we can do the following transformation (standardization) for $\hat{\beta}_1$: $\hat{\beta}_1 - \beta_1 \sim N\left(0, \sigma^2_{\hat{\beta}_1}\right)$, and then

$$\frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0, 1), \tag{6}$$

where $\sigma_{\hat{\beta}_1}$ denotes the standard deviation of $\hat{\beta}_1$. Notice that the standard deviation is the square root of the variance. (Since we use $\sigma^2_{\hat{\beta}_1}$ to denote the variance of $\hat{\beta}_1$, we use $\sigma_{\hat{\beta}_1}$ to denote its standard deviation.)
Now, we want to show that under $H_0 : \beta_1 = 1$ (that is, when H_0 is true), the t-statistic also follows $N(0, 1)$ under large samples. Note that the following decomposition always

holds:

$$t = \frac{\hat{\beta}_1 - 1}{SE(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} + \frac{\beta_1 - 1}{SE(\hat{\beta}_1)}. \tag{7}$$

When H_0 is true, $\beta_1 = 1$ and $\frac{\beta_1 - 1}{SE(\hat{\beta}_1)} = 0$. Therefore, when H_0 is true,

$$t = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)}. \tag{8}$$

Furthermore, we note that the standard error $SE(\hat{\beta}_1)$ is the estimator of $\sigma_{\hat{\beta}_1}$, so under large samples, $SE(\hat{\beta}_1)$ will be very close to $\sigma_{\hat{\beta}_1}$ (thus we can replace $\sigma_{\hat{\beta}_1}$ with $SE(\hat{\beta}_1)$). Combining this fact with equations (6) and (8), we conclude that under large samples (when the OLS assumptions are satisfied), and when H_0 is true,

$$t = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \sim \frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0, 1). \tag{9}$$

3. (a) By the definition of the (two-sided) t-test, the zero null hypothesis for the slope is rejected while the zero null hypothesis for the intercept is not rejected at the 5% level:

$$|t_{\beta_1}| = \left| \frac{2.4 - 0}{1.2} \right| = 2 > 1.96, \ |t_{\beta_0}| = \left| \frac{1.5 - 0}{1} \right| = 1.5 < 1.96. \tag{10}$$

   (b) As X actually has no effect on Y, $H_0 : \beta_1 = 0$ is true. Based on the definition of the 10% significance level (tolerance level for Type I error), the t-test has Type I error equal to 10%: even when H_0 is true, the t-test still can reject H_0 10% of the time: the probability of rejecting H_0 even when H_0 is true is equal to 10% (that is, $P_{H_0}(\text{Reject } H_0) = 10\%$). Now since 2,000 random samples are repeatedly drawn from the population, we expect that among these samples the t-test will reject $2,000 \times 10\% = 200$ of them.

4. (a) First, we have

$$E[e_i | income_i] = E[e_i] = 0, \tag{11}$$

   where the first equality follows from the independence between e_i and $income_i$.
   Second, to show the OLS Assumption 1 is satisfied, we note that

$$E[u_i | income_i] = E[\sqrt{income_i} \cdot e_i | income_i] = \sqrt{income_i} \cdot E[e_i | income_i] = 0, \tag{12}$$

   where the second equality follows from the property of conditional expectation and the third equality follows from (11). The conclusion follows.

   (b) Given that the intercept is equal to zero, the OLS minimization problem (only with respect to b_1 in this case) becomes:

$$\min_{b_1} \sum_{i=1}^{n} (Y_i - b_1 X_i)^2, \tag{13}$$

whose derivative with respect to b_1 is equal to $-2\sum_{i=1}^{n}(Y_i - b_1 X_i)X_i$. Setting this to zero and solving for the OLS estimator of the slope/gradient (i.e., we obtain the following FOC for $\hat{\beta}_1$):

$$-2\sum_{i=1}^{n}(Y_i - \hat{\beta}_1 X_i)X_i = 0. \tag{14}$$

Solving the equation, we find that

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} X_i Y_i}{\sum_{i=1}^{n} X_i^2}. \tag{15}$$